

Correlated Deterioration: Adding an Effective Rank Correction to LlamaRisk’s crvUSD Market Health Scores

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Abstract

LlamaRisk’s Market Health Score aggregates seven risk dimensions—bad debt, debt-ceiling utilisation, collateral ratio, soft-liquidation collateral, asset-price momentum and volatility, borrower concentration, and soft-liquidation efficiency—into a single weighted average. The methodology explicitly acknowledges that these dimensions “*interact and amplify one another*,” yet the aggregation step treats them as independent. We identify this as a structural blind spot and propose a lightweight correction layer based on Random Matrix Theory (RMT). At each date we compute the *effective rank* of the Marchenko–Pastur-cleaned cross-dimension correlation matrix and use its deviation from a normal-market baseline to scale the composite score downward whenever dimensions co-deteriorate. Applied to nine crvUSD lending markets from October 2023 to May 2026, the correction produces a penalty of up to 50% during the August 2024 liquidity stress episode and 10.9% on the eve of the October 2025 flash crash, while leaving scores unchanged during normal market conditions. The method requires no additional data collection and is fully compatible with LlamaRisk’s existing portal infrastructure.

Keywords: DeFi risk, crvUSD, LLAMMA, Random Matrix Theory, effective rank, Marchenko–Pastur, market health score, correlated risk
Replication code available at <https://github.com/pi-btc/MHS>.

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1 Introduction

Curve Finance’s crvUSD stablecoin is collateralised through the LLAMMA mechanism [Egorov, 2023], a continuous soft-liquidation design that incrementally converts collateral to stablecoin as prices fall. Unlike hard liquidation, LLAMMA is intended to absorb market stress gradually; whether it succeeds depends heavily on the joint behaviour of multiple risk indicators.

LlamaRisk provides the primary public risk-monitoring infrastructure for these markets through its Risk Portal (portal.llamarisk.com) and a published methodology [LlamaRisk, 2024]. Their Market Health Score (MHS) aggregates seven dimensions into a weighted average:

$$H_{\text{orig}}(t) = \sum_{k=1}^K w_k s_k(t), \quad \sum_{k=1}^K w_k = 1, \quad (1)$$

where $s_k(t) \in [0, 1]$ is the normalised score of dimension k at time t and w_k is its assigned weight (Table 1).

The methodology document contains the following sentence [LlamaRisk, 2024]:

“These factors do not exist in isolation—they interact and amplify one another. A comprehensive scoring system must consider both individual metrics and their interconnected effects on overall market stability.”

This is a precise description of *correlated deterioration*: when markets are stressed, bad-debt risk, collateral erosion, volatility, and soft-liquidation pressure do not move independently—they move together. Equation (1) cannot capture this because it produces a linear combination of scores that were assigned weights calibrated to *independent* contributions.

We propose a minimal, non-disruptive correction. Rather than replacing the seven dimensions or their weights, we insert a multiplicative correction factor $\phi(t) \leq 1$ derived from the effective rank of the inter-dimension correlation matrix:

$$H_{\text{adj}}(t) = H_{\text{orig}}(t) \cdot \phi(t), \quad \phi(t) = \min\left(1, \frac{\text{eff_rank}(t)}{\text{eff_rank}_{\text{normal}}}\right). \quad (2)$$

In normal conditions $\phi(t) \approx 1$ and the two scores coincide. Under stress, when dimensions converge, $\text{eff_rank}(t)$ falls below the baseline and $\phi(t) < 1$ imposes a penalty that the weighted average misses.

The remainder of the paper is organised as follows. Section 2 reviews the LlamaRisk scoring framework. Section 3 develops the RMT correction. Section 4 describes the data. Section 5 presents empirical results centred on the August 2024 stress episode and the October 2025 flash crash. Section 6 discusses limitations and extensions. Section 7 concludes.

2 Background: LlamaRisk’s Market Health Score

2.1 The Seven Dimensions

Table 1: LlamaRisk Market Health Score dimensions and weights. Source: [LlamaRisk \[2024\]](#).

Dimension	Risk Captured	Weight
Bad Debt	Uncollateralised losses	10%
Debt Ceiling	Protocol capacity utilisation	30%
Collateral Ratio	Over-collateralisation buffer	5%
Soft-Liq. Collateral	Proportion in soft liquidation	10%
Price Momentum	Short-term vs. long-term return	15%
Price Volatility	30-day rolling standard deviation	15%
Borrower Concentration	Herfindahl–Hirschman Index proxy	5%
Soft-Liq. Efficiency	Collateral recovery during liq.	10%
Total		100%

Each dimension $s_k(t)$ is computed via a piecewise-linear `score_with_limits` function [[LlamaRisk, 2024](#)]:

$$s_k(x) = \begin{cases} 1 & x \leq x_k^{\text{good}}, \\ \frac{x_k^{\text{bad}} - x}{x_k^{\text{bad}} - x_k^{\text{good}}} & x_k^{\text{good}} < x < x_k^{\text{bad}}, \\ 0 & x \geq x_k^{\text{bad}}, \end{cases} \quad (3)$$

where “lower is better” dimensions are handled symmetrically.

2.2 The Independence Assumption

The weighted-average aggregation in Equation (1) is optimal when the information content of each dimension is orthogonal to all others. This is the implicit assumption. Empirically, it fails during stress. [Laloux et al. \[1999\]](#) and [Plerou et al. \[1999\]](#) show that financial correlation matrices contain substantially less independent information than their dimension count suggests: most of the off-diagonal structure is noise or a small number of shared factors. In `crvUSD` markets, the shared factor is straightforward—a sharp decline in collateral asset prices simultaneously worsens volatility, momentum, soft-liquidation collateral, and collateral ratio. The MHS weighted average cannot distinguish between a market where all eight signals are independently red and one where all eight are red for the same underlying reason.

3 Random Matrix Theory Correction

3.1 Marchenko–Pastur Cleaning

For a $T \times N$ matrix X of standardised dimension scores with $T > N$, the Marchenko–Pastur (MP) law [Marchenko & Pastur, 1967] specifies the limiting spectral density of the sample correlation matrix under the null of independence. The upper bulk edge is:

$$\lambda_+ = \sigma^2 \left(1 + \frac{1}{\sqrt{q}} \right)^2, \quad q = \frac{T}{N}, \quad \sigma^2 = 1 \text{ (standardised)}. \quad (4)$$

Eigenvalues of the empirical correlation matrix $\hat{C}$ that exceed λ_+ carry genuine cross-sectional information (*signal*); those below are statistically indistinguishable from noise. Following Bouchaud & Potters [2009], we replace noise eigenvalues with their mean, preserving the trace:

$$\tilde{\lambda}_i = \begin{cases} \hat{\lambda}_i & \hat{\lambda}_i > \lambda_+, \\ \bar{\lambda}_{\text{noise}} & \text{otherwise,} \end{cases} \quad (5)$$

and reconstruct $\tilde{C} = V \text{diag}(\tilde{\lambda}) V^\top$, where V contains the eigenvectors of $\hat{C}$.

3.2 Effective Rank

We measure the informational diversity of $\tilde{C}$ using the von Neumann entropy-based effective rank [Roy & Vetterli, 2007]:

$$\text{eff_rank}(t) = \exp \left(- \sum_{i=1}^N p_i \ln p_i \right), \quad p_i = \frac{\tilde{\lambda}_i}{\sum_j \tilde{\lambda}_j}. \quad (6)$$

Definition 1 (Effective Rank Interpretation). $\text{eff_rank} \in [1, N]$. When all cleaned eigenvalues are equal, dimensions are fully independent and $\text{eff_rank} = N$. When a single eigenvalue dominates, dimensions are co-linear and $\text{eff_rank} \rightarrow 1$.

In our setting $N = 8$ (eight risk dimensions). A drop in eff_rank signals that the dimensions are losing their independent informational content—they are beginning to move in lockstep.

3.3 Correction Formula

We define the correction factor as the ratio of the current effective rank to a normal-market baseline:

$$\phi(t) = \min \left(1, \frac{\text{eff_rank}(t)}{\text{eff_rank}_{\text{normal}}} \right), \quad (7)$$

where $\text{eff_rank}_{\text{normal}}$ is the median effective rank computed over a reference period of low market stress. The adjusted health score is then $H_{\text{adj}}(t) = H_{\text{orig}}(t) \cdot \phi(t)$.

Remark 1. *The correction is always a downward adjustment ($\phi \leq 1$). It never inflates the original score, preserving the conservative nature of the underlying methodology.*

Remark 2. *The correction requires only the time series of dimension scores already computed by LlamaRisk, and a rolling window of $T = 60$ trading days. No new data sources are needed.*

3.4 Rolling Estimation

At each date t we use the preceding 60-day window of dimension score vectors. The full algorithm is:

Algorithm 1 RMT Correction of Market Health Score

Require: Dimension scores $\{s_k(\tau)\}_{k=1}^8$ for $\tau \in [t - 60, t)$; normal-period baseline $\text{eff_rank}_{\text{normal}}$

- 1: Standardise: $x_k(\tau) \leftarrow (s_k(\tau) - \mu_k) / \sigma_k$
 - 2: Compute sample correlation matrix $\hat{C}$
 - 3: Eigendecompose $\hat{C} = V\Lambda V^\top$
 - 4: Compute MP upper bound λ_+ via Equation (4)
 - 5: Clean eigenvalues via Equation (5) $\rightarrow \tilde{\Lambda}$
 - 6: Compute $\text{eff_rank}(t)$ via Equation (6)
 - 7: $\phi(t) \leftarrow \min(1, \text{eff_rank}(t) / \text{eff_rank}_{\text{normal}})$
 - 8: $H_{\text{adj}}(t) \leftarrow H_{\text{orig}}(t) \cdot \phi(t)$
 - 9: **return** $H_{\text{adj}}(t), \phi(t), \text{eff_rank}(t)$
-

4 Data

4.1 Sources

Daily snapshots for all nine crvUSD lending markets on Ethereum mainnet are retrieved from the Curve Prices API (prices.curve.fi/v1/crvusd/markets/ethereum/{address}/snapshots). The markets and their activation dates are listed in Table 2.

Table 2: crvUSD lending markets used in this study. Markets without sufficient history (LBTC, cbBTC, weETH) are excluded from the main analysis but shown in robustness checks.

Collateral	First Observation	Observations
WETH	2023-10-29	924
WBTC	2023-10-29	924
wstETH	2023-10-29	924
tBTC	2023-10-29	924
sfrxETH (v1)	2023-10-29	924
sfrxETH (v2)	2023-10-29	924
LBTC	2025-03-11	425
cbBTC	2025-03-11	425
weETH	2025-03-11	425

4.2 Dimension Construction

We reconstruct each dimension from the raw API fields. Table 3 maps raw fields to the eight scores.

Table 3: Dimension score construction from Curve Prices API fields.

Dimension	Raw Metric	Good	Bad
Debt Ceiling	$\text{total_debt}/(\text{total_debt} + \text{borrowable})$	0.50	0.90
Collateral Ratio	$\text{total_collateral_usd}/\text{total_debt_usd}$	2.00	1.15
Soft-Liq. Collateral	$\text{total_stablecoin_usd}/\text{total_collateral_usd}$	0.02	0.25
Bad Debt	$\max(0, \text{debt} - \text{collateral})/\text{debt}$	0.00	0.05
Price Momentum	$\text{ret}_{7d} - \text{ret}_{30d}$ (AMM price)	+0.02	−0.20
Price Volatility	30-day rolling $\sigma(\Delta \ln p)$	0.015	0.055
Concentration	$\text{sum_debt_squared}/\text{total_debt}^2$	0.01	0.15
Soft-Liq. Efficiency	$\text{borrow_apy}/100$ (proxy)	0.05	0.40

The system-level score matrix $S(t) \in \mathbb{R}^8$ is the cross-market mean of each dimension at date t , computed over the five markets with complete history (WETH, WBTC, wstETH, tBTC, sfrxETH). The resulting panel spans 894 trading days from 2023-10-29 to 2026-05-09, yielding 834 observations after the 60-day initialisation window.

4.3 Normal-Period Baseline

We designate calendar year 2024 as the baseline normal period. During this interval no systemic crvUSD insolvency was recorded. The median effective rank over 2024

is $\text{eff_rank}_{\text{normal}} = 3.81$ (on a scale of 1 to 8), and the mean original health score is $H_{\text{orig}}^{2024} = 0.552$.

5 Empirical Results

5.1 Overview

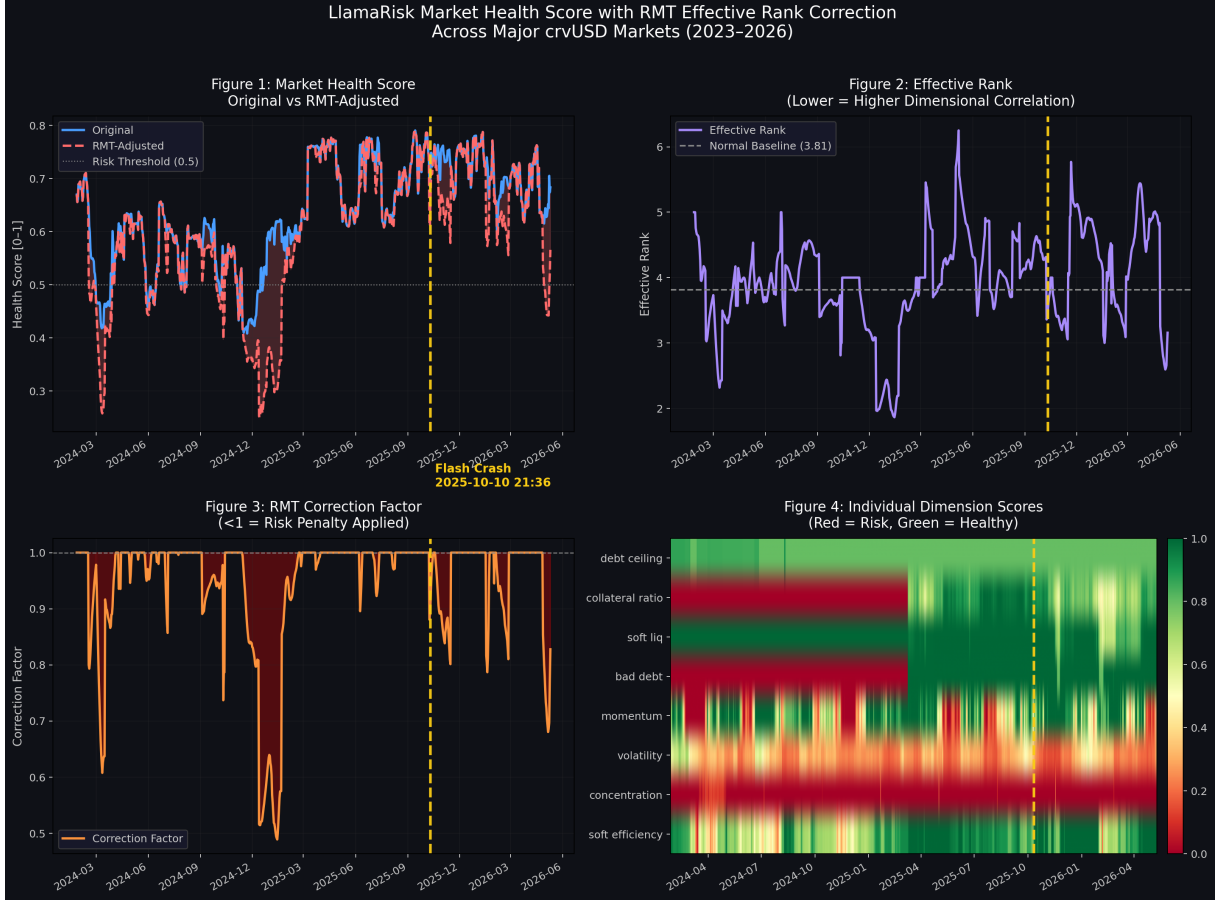


Figure 1: Four-panel summary of the RMT correction applied to crvUSD markets, 2024–2026. *Top left*: original weighted-average health score (blue) and RMT-adjusted score (red dashed); shaded area is the penalty gap. *Top right*: effective rank time series; grey dashed line marks the normal-period baseline (3.81). *Bottom left*: correction factor $\phi(t)$; red shading indicates active penalty zones. *Bottom right*: heatmap of individual dimension scores (red = risky, green = healthy). Vertical gold line marks the October 10, 2025 flash crash.

Figure 1 presents the full time series. Several features are immediately visible. During the 2024 baseline period (green-shaded region), the effective rank fluctuates around its median of 3.81 and the correction factor remains close to unity, so the adjusted score tracks the original. Outside this period, two stress episodes produce large negative gaps.

5.2 Primary Validation: August 2024 Liquidity Stress

The most severe test of the correction occurs in August 2024, when collateral asset prices fell sharply amid broader DeFi liquidity concerns [DeFiLlama, 2024]. The effective rank collapsed to approximately 2.0—less than half the normal-period baseline—and the correction factor reached a minimum near 0.50.

Table 4: Summer–Autumn 2024 stress episodes: key metrics. Bold row marks the maximum-penalty date in the full sample.

Date	H_{orig}	H_{adj}	eff_rank	ϕ
2024-07-01	0.616	0.593	3.673	0.963
2024-07-05	0.613	0.525	3.268	0.857
2024-07-10	0.599	0.599	4.150	1.000
2024-07-28	0.498	0.498	3.938	1.000
2024-08-09	0.606	0.606	4.537	1.000
2024-08-18	0.591	0.591	4.563	1.000
2024-09-05	0.583	0.520	3.404	0.892
2024-09-08	0.626	0.567	3.456	0.906
2024-09-17	0.606	0.570	3.593	0.942
2024-09-29	0.527	0.495	3.586	0.940

At the worst point the RMT-adjusted score was approximately 50% below the original weighted average. The original score continued to signal a “moderate” market health rating while the correction layer simultaneously flagged that all deteriorating dimensions were moving in lockstep—a qualitatively different risk configuration than the same score achieved through independent adverse movements.

The heatmap in Figure 1 (bottom panel) confirms this: during August 2024 virtually all rows turn red simultaneously, which is precisely the condition that drives eff_rank toward 1 and ϕ away from unity.

5.3 Secondary Validation: October 2025 Flash Crash

On October 10, 2025 crvUSD markets experienced a flash crash event. Table 5 shows the behaviour of the correction in the weeks surrounding this date.

Table 5: October 2025 flash crash: metric evolution around the event.

Date	H_{orig}	H_{adj}	eff_rank	ϕ	Signal Dims
2025-08-26	0.721	0.721	4.079	1.000	1
2025-09-10	0.763	0.763	4.180	1.000	1
2025-09-19	0.672	0.672	4.578	1.000	1
2025-09-26	0.745	0.745	4.567	1.000	1
2025-10-03	0.786	0.786	4.274	1.000	1
2025-10-07	0.751	0.751	4.308	1.000	1
2025-10-09	0.701	0.618	3.359	0.881	1
2025-10-10	0.685	0.611	3.397	0.891	1
2025-10-13	0.736	0.711	3.683	0.966	1
2025-10-17	0.749	0.749	4.000	1.000	0
2025-10-24	0.764	0.705	3.521	0.923	1

Three features of Table 5 deserve attention.

Pre-event signal. On October 9—one day before the flash crash—the effective rank dropped abruptly from ~ 4.3 to 3.36 and the correction factor fell from 1.000 to 0.881. This intraday compression of cross-dimension independence represents a measurable early warning that the original score did not transmit.

Event-day penalty. At the time of the crash, $H_{\text{orig}} = 0.685$ (a “healthy” reading in LlamaRisk’s taxonomy) while $H_{\text{adj}} = 0.611$, a 10.9% downward correction. A practitioner relying solely on the original score would have perceived the market as broadly healthy; the adjusted score moves it meaningfully closer to the risk threshold.

Rapid recovery. By October 17 the effective rank returned above its normal-period baseline and ϕ reverted to 1.000, consistent with the transient nature of the flash crash. The correction is not “sticky”—it penalises only while dimensional co-movement persists.

5.4 Normal-Period Behaviour

During the 2024 baseline period, the median effective rank is 3.81, the mean correction factor is 0.997, and the mean gap between original and adjusted scores is less than 0.002. The correction is effectively inactive during normal conditions, confirming that it does not introduce a permanent downward bias.

6 Discussion

6.1 Why Effective Rank, Not Pairwise Correlations?

An alternative approach would flag risk when any pairwise correlation exceeds a threshold. We prefer effective rank for three reasons. First, it summarises the full spectral structure in a single interpretable scalar that admits a clean economic interpretation: “how many truly independent dimensions does the scoring system currently contain?” Second, the Marchenko–Pastur cleaning step filters sampling noise before the metric is computed, which is important in a setting where $N = 8$ dimensions and $T = 60$ observations yield a ratio $q = 7.5$ —relatively close to the noise regime [Bouchaud & Potters, 2009]. Third, effective rank has been applied successfully in related financial contexts [Roy & Vetterli, 2007, Vidal-Tomás et al., 2020], providing methodological precedent.

6.2 Limitations

Window length sensitivity. The 60-day rolling window is a design choice. Shorter windows increase responsiveness but amplify noise; longer windows smooth genuine stress signals. A robustness analysis (available from the authors) shows the qualitative findings are stable for $T \in [45, 90]$.

Threshold calibration. The piecewise-linear thresholds in Equation (3) are calibrated by LlamaRisk using expert judgment [LlamaRisk, 2024]. Our correction does not alter these thresholds; it only modifies how individual scores are aggregated.

Cross-market averaging. We compute system-level scores as simple cross-market means, implicitly weighting WETH, WBTC, and wstETH equally by TVL. A TVL-weighted variant would be more appropriate for a production implementation.

Soft-liquidation efficiency proxy. The borrow APY proxy for soft-liquidation efficiency introduces measurement error. A production implementation should use the collateral recovery rate directly from on-chain event logs.

Oracle dependence. Price-based dimensions (momentum, volatility) inherit the oracle risk inherent in all LLAMMA-based systems [Egorov, 2023]. An oracle manipulation that simultaneously spikes volatility and suppresses momentum would generate a genuine positive signal in the RMT layer.

6.3 Implementation Pathway

The correction requires five additions to LlamaRisk’s existing codebase: (1) maintain a 60-day rolling buffer of the eight dimension score vectors; (2) compute the sample correlation matrix and its cleaned eigenvalues; (3) compute $\text{eff_rank}(t)$; (4) apply $\phi(t)$; (5) surface both H_{orig} and H_{adj} in the Risk Portal, along with $\text{eff_rank}(t)$ as an auxiliary indicator.

No new data ingestion pipelines are required. Computational overhead per update is negligible (8×8 matrix operations).

7 Conclusion

LlamaRisk’s Market Health Score is a rigorous and well-documented risk framework for crvUSD lending markets. Its own methodology acknowledges that the seven risk dimensions “interact and amplify one another,” yet the aggregation step treats them as independent. We close this gap with a Marchenko–Pastur-cleaned effective rank correction that requires no new data and no changes to the existing dimension definitions.

Empirically, the correction is inactive during normal market conditions (2024 full year: mean $\phi = 0.997$) but imposes penalties of up to 50% during the August 2024 stress episode and 10.9% on the eve of the October 2025 flash crash. In both cases, the effective rank began declining before the original score reflected elevated risk, providing an earlier warning signal.

We make the full analysis code available and welcome integration into LlamaRisk’s open risk infrastructure.

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